

Fedora Workstation 43 — Complete System Setup Log

User: rahulbagul

Install Type: Clean Install (NVMe SSD + HDD)

Desktop: GNOME (Wayland)

Shell: bash

Date: Jan 2026

Status: **STABLE · VERIFIED · PRODUCTION READY**

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BASE SYSTEM

- Fedora Workstation 43 (Clean Install)
- GNOME Desktop on Wayland
- Default shell: bash
- No zsh / no oh-my-zsh
- No PATH modifications

Goal:

Stable, predictable, professional Linux workstation.

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STORAGE SETUP (HDD AUTO-MOUNT)

Disk Layout

SSD — System Disk

/dev/nvme0n1

├─ nvme0n1p1 → EFI

├─ nvme0n1p2 → /boot

└─ nvme0n1p3 → / (root + home)

HDD — Data Disk

/dev/sda

├─ sda1 → ext4 → LABEL=Code → UUID=9cdd6ed9-d124-4a0a-9ea2-eb9aefde82f8

├─ sda2 → ext4 → LABEL=Temp → UUID=5858e361-d76f-4fde-ae7f-93c025274dd7

└─ sda3 → ext4 → LABEL=DATA → UUID=9c8f550a-435c-4897-9db7-8d5925b747bb

Mount Points

```
sudo mkdir -p /mnt/code
sudo mkdir -p /mnt/temp
sudo mkdir -p /mnt/data
```

/etc/fstab

```
UUID=9cdd6ed9-d124-4a0a-9ea2-eb9aefde82f8 /mnt/code ext4 defaults,x-gvfs-
show 0 2
UUID=5858e361-d76f-4fde-ae7f-93c025274dd7 /mnt/temp ext4 defaults,x-gvfs-
show 0 2
UUID=9c8f550a-435c-4897-9db7-8d5925b747bb /mnt/data ext4 defaults,x-gvfs-
show 0 2
```

Apply & permissions:

```
sudo systemctl daemon-reload
sudo mount -a
sudo chown -R rahulbagul:rahulbagul /mnt/code /mnt/temp /mnt/data
```

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DOWNLOADS FOLDER (UPDATED — HDD STORAGE + SAME PATH)

Requirement:

Downloads data physically HDD par store ho, lekin user ko `~/Downloads` hi dikhai de.

Solution:

Bind mount using `/mnt/data/Downloads` → `~/Downloads`

Why bind mount (not symlink): - Applications ko real directory milta hai - Permissions & sandbox friendly - GNOME / browsers / Flatpak safe - Path change nahi hota (`~/Downloads` same)

Setup Steps (One-Time)

```
# Create target directory on HDD
mkdir -p /mnt/data/Downloads

# Ensure home Downloads exists
mkdir -p /home/rahulbagul/Downloads
```

/etc/fstab Entry

```
/mnt/data/Downloads /home/rahulbagul/Downloads none bind,x-gvfs-show 0 0
```

Apply

```
sudo systemctl daemon-reload
sudo mount -a
```

Result:

- Files physically stored on HDD (/mnt/data/Downloads)
- Visible & usable as ~/Downloads
- No symlink, no path breakage

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MULTIMEDIA CODECS (RPM FUSION)

- RPM Fusion enabled
- ffmpeg-free replaced with full ffmpeg
- x264 / x265 / HEVC / HEIF
- GStreamer bad + ugly
- PipeWire aptX

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GNOME EXTENSIONS (USER-INSTALLED)

Enabled via GNOME Extensions app only: - Blur my Shell - Clipboard Indicator - Compiz Alike Magic Lamp Effect - Coverflow Alt-Tab - gTile - Impatience - Just Perfection - Native Window Placement - User Themes

Policy: - No Flatpak extensions - Extensions app only

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APPLICATIONS INSTALLED (VERIFIED)

System / Core

- Fedora Media Writer
- Settings
- GNOME Tweaks
- Extensions
- Boxes

- Maps
- Weather
- Clocks
- Contacts
- Document Viewer

Development & Networking

- Visual Studio Code
- Postman
- Dev Toolbox
- Git
- GCC / Make
- Htop
- What IP

Multimedia

- VLC Media Player
- GNOME Video Player
- Camera
- PsTube
- Clapper

Graphics & Images

- Inkscape
- GIMP
- Upscayl
- Image Viewer

Office & Documents

- LibreOffice (Writer, Calc, Impress)
- ONLYOFFICE
- PDF Arranger
- Warehouse

Utilities & Productivity

- AppFlowy
- Apostrophe
- Planify
- Resources
- Save Desktop
- Switcheroo
- Mission Center
- Logged Devices
- Concessio

- Buzz
- Brief
- Pin It!
- Characters
- Gaphor
- Eyedropper

DEVELOPMENT TOOLS

- VS Code (DNF)
- Git
- GCC / Make
- Brave Browser (DNF)
- No Flatpak dev tools

NODE.JS ENVIRONMENT

- System-wide Node.js v22.x
- Installed via Fedora repos
- No NVM
- Clean PATH

SHELL PROMPT (STARSHIP)

- Installed via COPR (atim/starship)
- Binary: /usr/bin/starship
- Config: ~/.config/starship.toml
- No PATH hacks

DISPLAY POWER BEHAVIOR

```
gsettings set org.gnome.desktop.session idle-delay 0
```

- No auto screen-off

SECURITY & FIREWALL

```
sudo systemctl enable --now firewalld  
firewall-cmd --state
```

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SYSTEM SNAPSHOTS (TIMESHIFT)

- Mode: RSYNC
- /mnt excluded
- Restore SELinux fix:

```
touch /.autorelabel
```

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BACKUP STRATEGY

```
rsync -aAXHv --delete /home/rahulbagul /mnt/data/backup/home
```

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WI-FI 5 GHz LOCK

```
nmcli con modify "SSID" wifi.band a  
nmcli con modify "SSID" wifi.bssid AA:CA:E7:B8:E9:92  
nmcli con down "SSID"  
nmcli con up "SSID"
```

Verify:

```
iw dev wlp5s0 link
```

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MAINTENANCE COMMANDS

```
sudo dnf upgrade --refresh
sudo dnf autoremove
sudo dnf clean packages
sudo journalctl --vacuum-time=7d
```

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FINAL STATUS

- ✓ Clean Fedora 43 install
- ✓ Stable storage layout
- ✓ Verified apps & extensions
- ✓ GNOME tuned & locked
- ✓ Production-ready system

Document Generated: \$(date)

Hostname: \$(hostname)

Kernel: \$(uname -r)

Uptime: \$(uptime -p)